Labsheet-6

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Roll No-28

Secti on- B

1. Select a suitable dataset such as the House Price Prediction Dataset, Advertising Sales Dataset, or California Housing Dataset.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_scor
                                                                        scorc


housing = fetch_california_housing(as_frame=True)

df = housing.frame

print(df.head())
```

```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude  MedHouseVal
0    -122.23        4.526
1    -122.22        3.585
2    -122.24        3.521
3    -122.25        3.413
4    -122.25        3.422
```

MedInc HouseAge AveRooms AveBedrms Population AveOccup Latitude
e 8.3252 41.0 6.984127 1.023810 322.0 2.555556 37.88

1 8.3014 21.0 6.238137 o . 971880 2401. o 2.109842 37.86

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Longitude MedHouseVa1
o -122.23 4. 526

1 -122.22 3. 585

2	-122.24	3. 521
3	-122.25	3.413
4	-122.25	3.422

2. Import the required libraries including Pandas, NumPy, Matplotlib, and Scikit-learn, and load the dataset.

```
print("Shape of dataset:", df.shape)


print("\nColumns:")
print(df.columns)
```

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```
print("\nDataset Information:")
print(df.info())


print("\nMissing Values:")
print(df.isnull().sum())


print("\nStatistical Summary:")
print(df.describe())
```

3. Perform basic data preprocessing by handling missing values, encoding categorical variables (if any), and selecting relevant features.

```
X = df.drop("MedHouseVal", axis=1)
```

```
print("Missing values before handling:")
print(df.isnull().sum())


df = df.fillna(df.median(numeric_only=True))

print("\nMissing values after handling:")
print(df.isnull().sum())
```

```
Missing values before handling:
MedInc         0
HouseAge       0
AveRooms       0
AveBedrms      0
Population     0
AveOccup       0
Latitude       0
Longitude      0
MedHouseVal    0
dtype: int64

Missing values after handling:
MedInc         0
HouseAge       0
AveRooms       0
AveBedrms      0
Population     0
AveOccup       0
Latitude       0
Longitude      0
MedHouseVal    0
dtype: int64
```

4. Split the dataset into training and testing sets using an appropriate train-test ratio (e . g., 80:20).

```
y = df["MedHouseVal"]

print("Features:")
print(X.head())

print("\nTarget:")
print(y.head())
```

```
Features:
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude
0    -122.23
1    -122.22
2    -122.24
3    -122.25
4    -122.25

Target:
0    4.526
1    3.585
2    3.521
3    3.413
4    3.422
Name: MedHouseVal, dtype: float64
```

5. Train a Linear Regression model using the training dataset.

```
X_train, X_test, y_train, y_test -
    train_test_split( x,

    test          size=0.2Ø,
    random state=42



print ("Training data: " , X_train . shape)
print ("Testing data: " , X_test . shape)
```

Training data: (16512, 8)

Testing data: (4128, 8)

6. Use the trained model to predict values for the testing dataset.

```
model = LinearRegression()


model.fit(X_train, y_train)
```

```
print("Model training completed.")
```

```
Model training completed.
```

7. Visualize the regression line and compare the actual and predicted values using appropriate plots

```
y_pred = model.predict(X_test)

print("Actual values:")
print(y_test.head())

print("\nPredicted values:")
print(y_pred[:5])
```

```
Actual values:
20046    0.47700
3024     0.45800
15663    5.00001
20484    2.18600
9814     2.78000
Name: MedHouseVal, dtype: float64

Predicted values:
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725]
```

Evaluate the model using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

```
plt.figure(figsize=(8, 6))

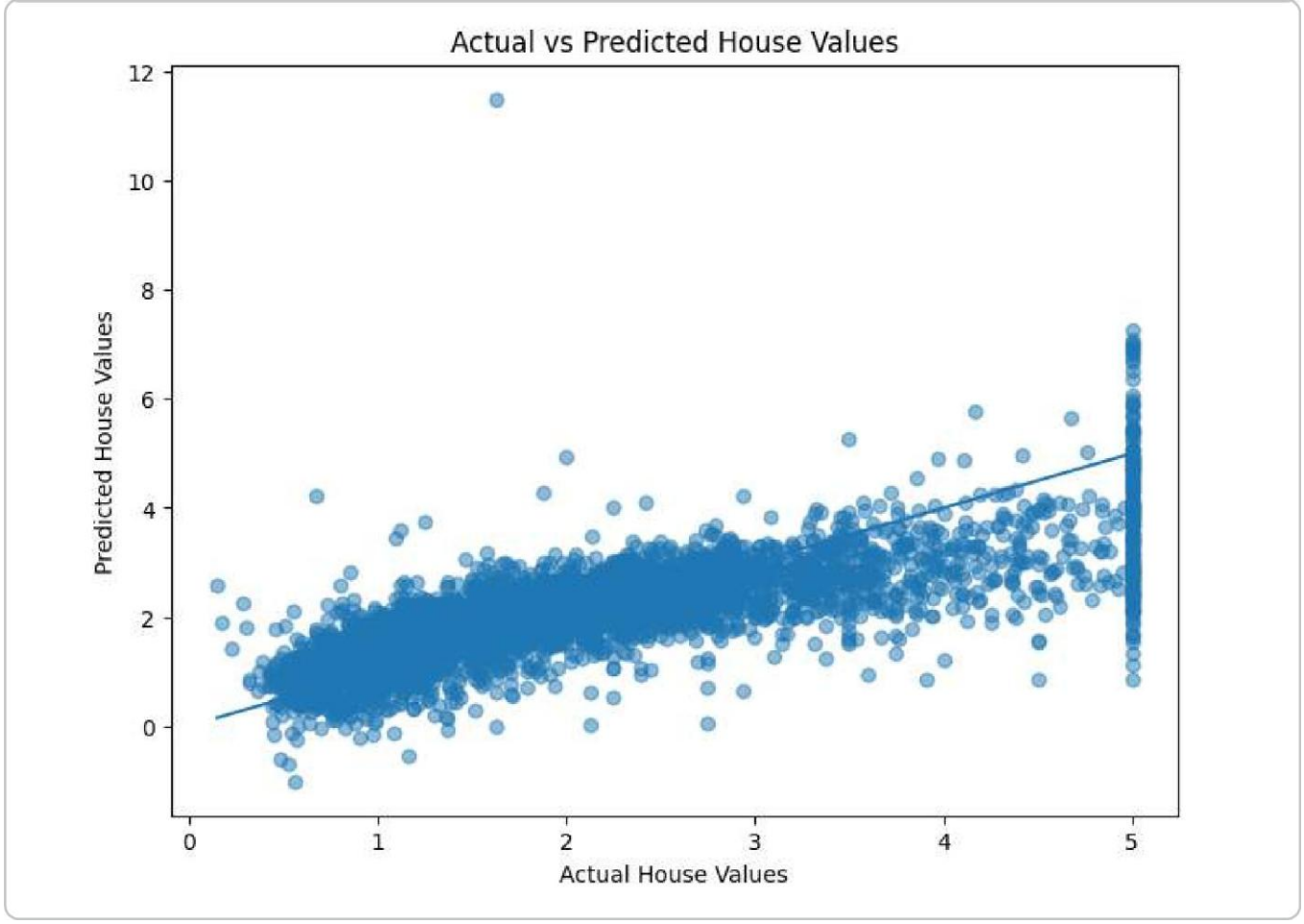
plt.scatter(y_test, y_pred, alpha=0.5)


plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()]
)

plt.xlabel("Actual House Values")
plt.ylabel("Predicted House Values")
plt.title("Actual vs Predicted House Values")

plt.show()
```

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9. Analyze the regression coefficients to understand the impact of each independent variable on the target variable.

```
mae = mean_absolute_error(y_test, y_pred)


mse = mean_squared_error(y_test, y_pred)


rmse = np.sqrt(mse)


r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R² Score:", r2)
```

```
Mean Absolute Error (MAE): 0.5332001304956553
Mean Squared Error (MSE): 0.5558915986952444
Root Mean Squared Error (RMSE): 0.7455813830127764
R² Score: 0.5757877060324508
```

10. Interpret the results and discuss the suitability of Linear Regression for the selected prediction problem.

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```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})


coefficients["Absolute_Coefficient"] = coefficients["Coefficient"].abs()

coefficients = coefficients.sort_values(
    by="Absolute_Coefficient",
    ascending=False
)

print(coefficients)
```

```
      Feature  Coefficient  Absolute_Coefficient
3   AveBedrms     0.783145              0.783145
0      MedInc     0.448675              0.448675
7   Longitude    -0.433708              0.433708
6    Latitude    -0.419792              0.419792
2    AveRooms    -0.123323              0.123323
1    HouseAge     0.009724              0.009724
5    AveOccup    -0.003526              0.003526
4  Population    -0.000002              0.000002
```

Feature Coefficient Absolute Coefficient

3 AveBedrms e. 783145 e. 783145 e
Medlnc e. 448675 448675

7 Longitude -0.433708 0.433708

6 Latitude -0.419792 0.419792

2 AveRooms -0.123323 e. 123323 1 HouseAge o . 009724
o .009724

5 AveOccup -o . 003526 0.003526
0.000002

4 Population -0.000002